Two-particle interaction

$$\hat{H}_{int} = \frac{1}{2}\int d\bar{r}_1 d\bar{r}_2 \; V(\bar{r}_1-\bar{r}_2)\, \hat{\rho}(\bar{r}_1)\, \hat{\rho}(\bar{r}_2)$$

In fact, instead of $\hat{\rho}(\bar{r}_1)\hat{\rho}(\bar{r}_2)$ we should write

proper

$$\hat{\varphi}^+(\bar{r}_1)\,\hat{\varphi}^+(\bar{r}_2)\,\hat{\varphi}(\bar{r}_2)\,\hat{\varphi}(\bar{r}_1) = -\hat{\varphi}^+_1\,\underbrace{\hat{\varphi}^+_2\,\hat{\varphi}_1}_{\delta(\bar{r}_2-\bar{r}_1)\,-\,\hat{\varphi}_1\hat{\varphi}^+_2}\,\hat{\varphi}_2 =$$

$$= -\delta(\bar{r}_1-\bar{r}_2)\;\hat{\varphi}^+(r_1)\,\hat{\varphi}(r_2) + \underbrace{\hat{\varphi}^+(r_1)\,\hat{\varphi}(r_1)}_{\hat{\rho}(r_1)}\;\underbrace{\hat{\varphi}^+(r_2)\,\hat{\varphi}(r_2)}_{\hat{\rho}(r_2)}$$

which shows that proper form

$$\hat{H}^{(pr)}_{int} = \frac{1}{2}\int d\bar{r}_1 d\bar{r}_2 \; V(\bar{r}_1-\bar{r}_2)\;\hat{\varphi}^+(r_1)\,\hat{\varphi}^+(r_2)\,\hat{\varphi}(r_2)\,\hat{\varphi}(r_1)$$

and naive form

$$\hat{H}_{int} = \frac{1}{2}\int d\bar{r}_1 d\bar{r}_2 \; V(\bar{r}_1-\bar{r}_2)\;\hat{\varphi}^+(r_1)\,\hat{\varphi}(r_1)\;\hat{\varphi}^+(r_2)\,\hat{\varphi}(r_2)$$

differ only by

$$\hat{H}^{(pr)}_{int} - \hat{H}_{int} = \frac{1}{2}\int d\bar{r}_1 d\bar{r}_2 \; V(\bar{r}_1-\bar{r}_2)\left(-\delta(\bar{r}_1-\bar{r}_2)\;\hat{\varphi}^+(r_1)\,\hat{\varphi}(r_2)\right)$$

$$= -\frac{1}{2}\int d\vec{r}\; V(0)\;\hat{\varphi}^+(r)\,\hat{\varphi}(r).$$

This difference represents correction to the chemical potential. We thus can discuss key physics by using either of the two forms $\rightarrow$ let's try to think in terms of naive density expression.

In a system with interactions $\hat{H}_{int}$ perturbation $\delta\rho(r,t)$ created by external probe $H'(t) = \int dr\, \lambda(r,t)\, \hat{\rho}(r)$

~~[illegible]~~ serves as "external" source for other modes it couples to via interaction $\hat{H}_{int}$!

To find the response of such system we resort to mean-field like approximation (known also as RPA = random phase approx.)

So let $\rho(\vec{r}) = \bar{\rho}(\vec{r}) + \delta\rho(\vec{r},t)$

Here $\bar{\rho}(\vec{r}) = \hat{\rho}(\vec{r}) - \rho_0$ represents fluctuation in the absence of $H'(t)$,

then $\langle \bar{\rho}(\vec{r}) \rangle = 0$ (if $H'(t) = 0$).

$\delta\rho(\vec{r},t)$ is caused by $H'(t)$.

Then $H_{int} = \frac{1}{2} \int dr_1 dr_2\, V(r_1 - r_2) \left(\bar{\rho}(r_1) + \delta\rho(r_1,t)\right)\left(\bar{\rho}(r_2) + \delta\rho(r_2,t)\right)$

$$\cong \frac{1}{2} \int dr_1 dr_2\, V(r_1 - r_2) \Big\{ \bar{\rho}(r_1)\, \bar{\rho}(r_2) + \bar{\rho}(r_1)\, \delta\rho(r_2,t) + \delta\rho(r_1,t)\, \bar{\rho}(r_2) + \cancel{\delta\rho(r_1,t)\, \delta\rho(r_2,t)} \Big\}$$

assume that $\delta\rho \cdot \delta\rho$ is small and can be disregarded...

Now express everything back

$\delta\rho(r,t) = \rho(r) - \bar{\rho}(r)$

$$H_{int} \approx \frac{1}{2} \int dr_1 dr_2\, V(r_1 - r_2) \Big\{ -\bar{\rho}(r_1)\, \bar{\rho}(r_2) + \rho(r_2)\, \bar{\rho}(r_1) + \rho(r_1)\, \bar{\rho}(r_2) \Big\}$$

↙ mean-field

In this approximation $H_{int}^{MF} = \int d\bar{r}\, \rho(r) \underbrace{\int d\bar{r}'\, V(r-r')\, \bar{\rho}(r')}_{\text{average potential created by other particles}}$

This means that $H'(t) = \int dr\, \lambda(r,t)\, \hat{\rho}(r)$

is extended to $H^{MF} = \int dr \left(\gamma(r,t) + \lambda(r,t)\right) \hat{\rho}(r)$

where $\gamma(r,t) = \int dr'\, V(r-r')\, \bar{\rho}(r',t)$.

In linear response

$$\bar{\rho}(r,t) = \int dr_1\, dt_1\, \chi_0(r-r_1, t-t_1) \left(\gamma(r_1,t_1) + \lambda(r_1,t_1)\right).$$

Fourier transform everything

$$\bar{\rho}(\bar{q},\omega) = \chi_0(\bar{q},\omega)\left(\gamma(\bar{q},\omega) + \lambda(\bar{q},\omega)\right)$$

but $\gamma(\bar{q},\omega) = V(\bar{q})\, \bar{\rho}(\bar{q},\omega)$

$$\Rightarrow \bar{\rho}(\bar{q},\omega) = \chi_0(\bar{q},\omega)\left(\lambda(\bar{q},\omega) + V(\bar{q})\, \bar{\rho}(\bar{q},\omega)\right)$$

$$\bar{\rho}(\bar{q},\omega)\left[1 - V(\bar{q})\, \chi_0(\bar{q},\omega)\right] = \chi_0(\bar{q},\omega)\, \lambda(\bar{q},\omega)$$

$$\bar{\rho}(\bar{q},\omega) = \frac{\chi_0(\bar{q},\omega)}{1 - V(\bar{q})\, \chi_0(\bar{q},\omega)}\, \lambda(\bar{q},\omega) = \chi(\bar{q},\omega)\, \lambda(\bar{q},\omega)$$

Susceptibility of interacting system $\chi(\bar{q},\omega) = \dfrac{\chi_0(\bar{q},\omega)}{1 - V(\bar{q})\, \chi_0(\bar{q},\omega)}$.

$\chi_0(\bar{q},\omega)$ is susceptibility of non-interacting Fermi gas.

Thus, interacting system responds with χ.
while non-interacting — with χ_0.

$\Rightarrow$ (1) Interactions renormalize response.

(2) One now has possibility of pole $1 - V(\bar{q})\chi_0(\bar{q},\omega) = 0$ which means that <u>collective excitations</u> are possible:

$$\left[1 - V(\bar{q})\chi_0(\bar{q},\omega)\right]\bar{\rho}(\bar{q},\omega) = \chi_0(\bar{q},\omega)\,\lambda(\bar{q},\omega):$$

If $\lambda(\bar{q},\omega) = 0$ we now have two options

(a) $\bar{\rho}(\bar{q},\omega) = 0 \rightarrow$ meaning, nothing happens.

(b) $\bar{\rho}(\bar{q},\omega_q) \neq 0$ because $V(\bar{q})\chi_0(\bar{q},\omega_q) = 1$

so that

$$\left[1 - V(\bar{q})\chi_0(\bar{q},\omega_q)\right]\bar{\rho}(\bar{q},\omega_q) = 0$$

is satisfied because expression in square brackets is <u>zero</u>

$1 = V(\bar{q})\chi_0(\bar{q},\omega_q)$ is implicit equation for ω_q = dispersion of collective mode.

Little thinking tells that these are actually two conditions:

$$\begin{cases} V(\bar{q})\ \mathrm{Re}\left[\chi_0(\bar{q},\omega_q)\right] = 1 \\ \mathrm{Im}\left[\chi_0(\bar{q},\omega_q)\right] = 0 . \end{cases}$$

<u>Plasmons</u>: consider $\omega \gg v_F q$ in $\chi_0(\bar{q},\omega)$.

Then $\mathrm{Im}\left[\chi_0(\bar{q},\omega)\right] = 0$ and

$$\mathrm{Re}\,\chi_0(\bar{q},\omega) = N_0 \left\{ -1 + \frac{\omega}{2v_F q} \ln \frac{\omega(1+v_F q/\omega)}{\omega(1-v_F q/\omega)} \right\} =$$

$X \equiv \frac{v_F q}{\omega} \ll 1$

$$= N_0 \left\{ -1 + \frac{1}{2x} \ln \frac{1+x}{1-x} \right\}$$

$$\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \dots\right)$$

$$\Rightarrow \mathrm{Re}\,\chi_0 \cong N_0 \left\{ -1 + \frac{1}{2x}\left(2x + \frac{2}{3}x^3 + \frac{2}{5}x^5 + \dots\right) \right\}$$

$$= N_0 \left\{ -1 + 1 + \frac{1}{3}x^2 + \dots \right\} = \frac{1}{3} N_0 \left(\frac{v_F q}{\omega}\right)^2 .$$

Thus $V(\bar{q}) \cdot \frac{1}{3} N_0 \left(\frac{v_F q}{\omega_q}\right)^2 = 1$ is the condition.

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$$\frac{4\pi e^2}{q^2} \frac{N_0}{3} \frac{v_F^2 q^2}{\omega_q^2} = 1 \Rightarrow \omega_q^2 = \frac{4\pi e^2}{3} N_0 v_F^2 .$$

But $N_0 = \frac{m k_F}{2\pi^2}$, $v_F = \frac{k_F}{m} \Rightarrow \omega_q^2 = \frac{4\pi e^2}{3} \frac{k_F^3}{2\pi^2 m} = \frac{4\pi e^2 \rho_0}{m} =$ plasma frequency.

and $k_F^3 = 6\pi^2 \rho_0$, $\rho_0 = \frac{N}{V}$ = electron density.

Note that for the Coulomb interaction

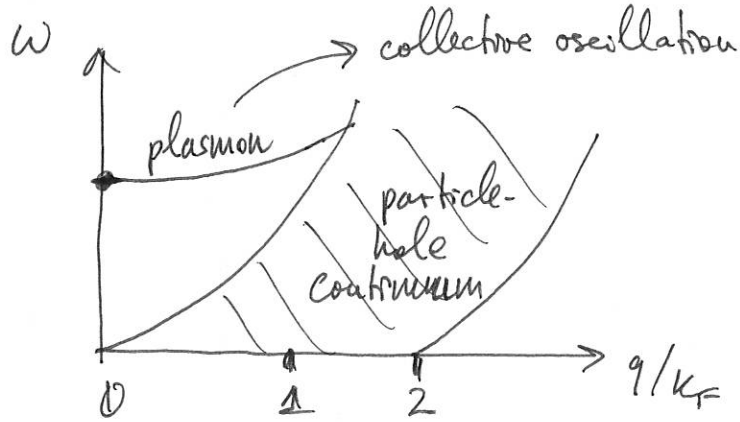
$$V(\vec{r}_1-\vec{r}_2) = V(\vec{r}) = \frac{e^2}{|\bar{r}_1-\bar{r}_2|} = \frac{e^2}{|\bar{r}|} = \frac{e^2}{r}$$

$$V(\bar{q}) = \int d\vec{r}\, V(\bar{r})\, e^{i\bar{q}\cdot\bar{r}} = \int_0^\infty dr\, r^2 \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta\, \frac{e^2}{r} e^{iqr\cos\theta}$$

$$= 2\pi e^2 \int_0^\infty dr\, r \int_{-1}^{1} d(\cos\theta)\, e^{iqr\cos\theta} = (\xi = \cos\theta) =$$

$$= 2\pi e^2 \int_0^\infty dr\, r \left. \frac{e^{iqr\xi}}{iqr} \right|_{-1}^{1} = 2\pi e^2 \int_0^\infty dr \cdot r\, \frac{2i\sin(qr)}{iqr} =$$

$$= \frac{4\pi e^2}{q} \int_0^\infty dr\, \sin(qr) = \frac{4\pi e^2}{q^2} \underbrace{\int_0^\infty d(qr)\, \sin(qr)}_{-\cos(qr)\big|_0^\infty = +1}$$

add regularization $e^{-\alpha r}$, and then set $\alpha = 0$ at the end.

$$V(\bar{q}) = \frac{4\pi e^2}{q^2}.$$